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Department of Applied Chemistry (2024-25)

NOTES

UNIT-IV

Drugs and Polymer Chemistry

Unit 4
Drugs and Polymer Chemistry

Drugs: Introduction, types of drugs, synthesis of commonly used drug molecules such as aspirin and paracetamol.

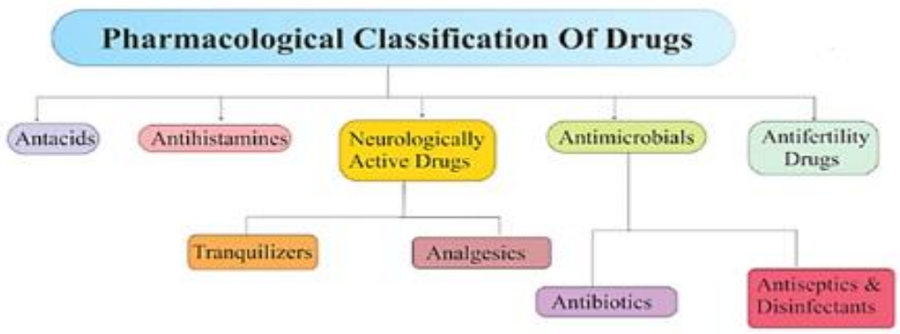
Introduction: A drug is a chemical substance, typically of known structure, which, when administered to a living organism, produces a biological effect. A pharmaceutical drug, also called a medication or medicine, is a chemical substance used to treat, cure, prevent, or diagnose a disease or to promote well-being.

Drugs are chemicals of low molecular masses (~100 - 500u). These interact with macromolecular targets and produce a biological response. When the biological response is therapeutic and useful, these chemicals are called medicines and are used in diagnosis, prevention and treatment of diseases. Over consumption of drugs can be lethal. Hence we can say that "All medicines are drugs, but all drugs are not medicines".

Types of Drugs
Medicinal science has discovered numerous drugs for various purposes.

Based on pharmacological effect
How a drug or medicine affects or influences the cells of an organism is referred to as the pharmacological effect. Different types of drugs have various pharmacological effects on an organism.
For example, an analgesic reduces pain while an anti-inflammatory drug reduces the inflammation of the body. Thus, drugs can be classified based on the pharmacological effect.

Different Types of Drugs Based on Pharmacological/Therapeutic Effects



A) Antacids
Antacid is a substance that reduces the level of acid in the stomach. Overproduction of acid in the stomach causes irritation, pain or even ulcers in the stomach. Antacids neutralize the excess of acid produced in the body. Thus, these drugs help to maintain the pH in the stomach. For example, ranitidine (Zantac), cimetidine (Tegamet) etc.

B) Antihistamines
Antihistamines are the drugs which have been used to prevent and control allergies. These drugs work against histamine. Some people suffer hypersensitivity to dust, due to release of histamine in their body. Antihistamine drugs are used for allergies, fever, sneezing etc. Some common drugs used as antihistamine are brompheniramine, cetirizine, and terfenadine.

C) Antipyretics
Antipyretics are used to bring down the body temperature during high fever. For example Aspirin, Phenacetin, Ibuprofen and Paracetamol.

D) Analgesic
These are drugs we commonly identify as painkillers. They mainly interact with elements of the nervous system. They reduce or extinguish any pain or discomfort felt anywhere in the body, but without impairing consciousness or causing incoordination. There are two basic types of analgesic

i) Non-narcotic Analgesic: Also known as the simple analgesic or Non-steroidal anti-inflammatory drugs (NSAIDs). These are medicines that are widely used to relieve pain, reduce inflammation, and bring down a high temperature. They mainly reduce the

production of a chemical known as Prostaglandins, which is normally produced in response to tissue damage. For example Aspirin, Phenacetin, Ibuprofen and Paracetamol.

ii) Narcotic Analgesics: Drugs which produce sleep and unconsciousness are called narcotics. These are helpful in moderate to severe pain management. They basically attach to the receptors in your brain and reduce the perception of the pain. Some can also increase the patient's threshold of pain. In high doses, they are very dangerous and addictive as well. Some examples include morphine, oxycodone, fentanyl etc.

E) Antimicrobials

These drugs inhibit the growth of disease-causing microorganisms in the body. Bacteria, viruses, fungi etc. causes many diseases in human beings and animals as well. For the treatment of pathogenic action of microbes in the body, many kinds of antimicrobial drugs have been discovered. For example, antibacterial drugs work against bacteria, antifungal drugs work against fungi, antiviral drugs work against viruses, antiparasitic drugs work against parasites selectively. Antibiotics, antiseptic and disinfectants are antimicrobial drugs.

F) Tranquilizers

These are chemically synthesized drugs that help humans fight a variety of mental diseases such as stress, anxiety, mental fatigue, depression etc. They are identified as neurologically active drugs. These drugs form an important component of sleeping pills. For example, noradrenaline is one of the neurotransmitters that play a vital role in mood changes. If the levels of noradrenaline get too low, a person may feel depressed. So drugs like chlordiazepoxide or Equanil inhibit the enzyme that is responsible for breaking down noradrenaline.

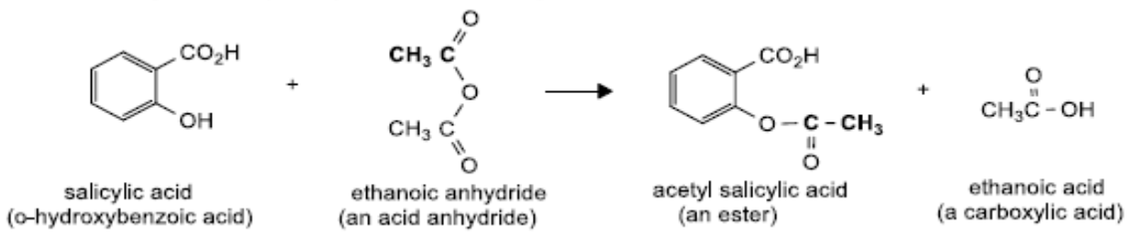
Synthesis of Commonly Used Drug Molecules

Synthesis of Aspirin (acetyl salicylic acid or ASA) Acetylsalicylic acid, commonly known as aspirin, is the most widely used drug in the world today. Aspirin is very powerful and effective analgesic (relieves pain), an antipyretic (reduces fever) and an anti-inflammatory (reduces swelling). Studies also suggest that aspirin can also reduce the risk of heart attack.

Salicylic acid, whose name comes from Salix, the willow family of plants, was derived from willow bark extracts. Hippocrates the ancient Greek physician, as well as Native Americans before Columbus' time, prepared willow bark teas as headache remedies and other tonics. In modern times, salicylic acid is administered in the form of aspirin which is less irritating to the stomach than salicylic acid. Aspirin was first synthesized in 1893 by Felix Hofmann, a chemist for the German firm of Bayer.

To prepare aspirin, salicylic acid is reacted with an excess of acetic anhydride. A small amount of a strong acid like sulfuric acid is used as the catalyst. Aspirin is prepared by reacting Salicylic acid (o-hydroxybenzoic acid) with ethanoic anhydride (acetic anhydride). This synthesis will involve the reaction of two functional groups, an -OH group of salicylic acid and an acid anhydride, to form the product, an ester (acetyl salicylic acid).

The most likely impurities in the crude acetyl salicylic acid are unreacted salicylic acid and a polymeric ester by-product. The polymeric by-product is not soluble in sodium bicarbonate and can be removed by dissolving the crude product in sodium bicarbonate, filtering to remove the insoluble polymer then acidifying to recover the acetyl salicylic acid. The ester product will be isolated, and purified by recrystallization using warm ethanol.



salicylic acid (o-hydroxybenzoic acid) | ethanoic anhydride (an acid anhydride) | acetyl salicylic acid (an ester) | ethanoic acid (a carboxylic acid)

Synthesis of Paracetamol (Acetaminophen)

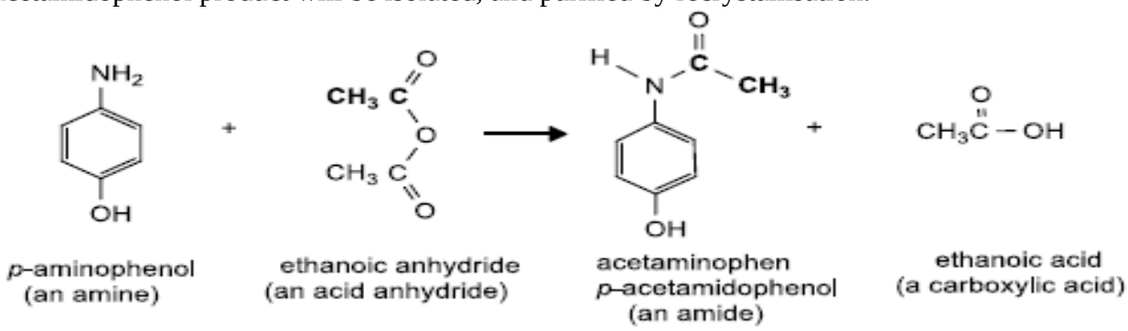
Paracetamol is the brand name for a medication that is commonly called acetaminophen, Tylenol or Panadol. Paracetamol is an analgesic (relieves pain) and an antipyretic (reduces fever).

Paracetamol was first synthesized in 1878 by Morse, by the reduction of p-nitrophenol with tin in glacial acetic acid and introduced for medical usage in 1883. It is the preferred alternative analgesic-antipyretic to aspirin, particularly in patients with coagulation disorders, individuals

with a history of peptic ulcer or who cannot tolerate aspirin, as well as in children.

A direct synthesis of paracetamol (acetaminophen) from hydroquinone has been developed using ammonium acetate as an amidating agent. The reaction proceeds in acetic acid at elevated temperatures without any metallic catalyst. Under these conditions, paracetamol was obtained with high yield and selectivity (>95%).

Paracetamol synthesis will involve the reaction of two functional groups, an amine of p-aminophenol and an acid anhydride to form a product p-acetamidophenol (Paracetamol). The p-acetamidophenol product will be isolated, and purified by recrystallisation.



Polymer: Introduction, Classification of polymers, Use and disposal of polymers. Properties of polymers - Solubility, Molecular Weight, Crystallinity, Glass transition temperature.

Introduction: A **polymer** is a large molecule or a macromolecule, which essentially is a combination of many subunits. The term polymer in Greek means 'many parts'. Polymers can be found all around us, from the strand of our DNA, which is a naturally occurring biopolymer, to polypropylene, polyethylene which is used throughout the world as plastic. Different polymers have a number of unique physical and chemical properties, due to which they find usage in everyday life.

Polymers can be naturally found in plants and animals (**natural polymers**) or can be human-made (**synthetic polymers**).

Polymers are created by the process of polymerization, wherein their constituent elements, called monomers, are reacted together to form polymer chains, i.e., 3-dimensional networks forming the polymer bonds.

Polymerization: Polymerization is a process through which a large number of monomer molecules react together to form a polymer.

Degree of polymerization (DP): It refers to the number of monomeric units in a polymer chain. Essentially, it indicates the chain length of the polymer and is a crucial factor in determining the polymer's physical properties, such as strength, viscosity, and thermal behavior

For a homopolymer, which consists of only one type of monomer, the degree of polymerization can be calculated using the formula:

$$\overline{DP}_n \equiv \overline{X}_n = \frac{\overline{M}_n}{M_0}$$

$\overline{M}_n$ is the number-average molecular weight

M_0 is the molecular weight of the monomer unit.

The overlines indicate arithmetic mean values.

A higher degree of polymerization generally means a higher molecular weight, which often correlates with enhanced mechanical properties and higher melting temperatures. For most industrial purposes, degrees of polymerization in the thousands or tens of thousands are desired. This number does not reflect the variation in molecule size of the polymer that typically occurs, it only represents the mean number of monomeric units.

Classification of Polymers

Polymers can be classified in different categories depending on their structures, different behaviours and vast applications.

Classification of Polymers Based on the Source of Availability

1. **Natural Polymers:** They occur naturally and are found in plants and animals. For example, proteins, starch, cellulose and rubber. To add up, we also have biodegradable polymers called biopolymers.
2. **Semi-synthetic Polymers:** They are derived from naturally occurring polymers and undergo further chemical modification. For example, cellulose nitrate and cellulose acetate.
3. **Synthetic Polymers:** These are human-made polymers. Plastic is the most common and widely used synthetic polymer. It is used in industries and various dairy products. For example, nylon-6, 6, polyether, etc.

Classification of Polymers Based on the Structure of the Monomer Chain

1. **Linear Polymers:** The structure of polymers containing long and straight chains falls into this category. PVC, i.e., polyvinyl chloride, is largely used for making pipes, and an electric cable is an example of a linear polymer.
2. **Branched-chain Polymers:** When linear chains of a polymer form branches, then such polymers are categorised as branched chain polymers. For example, low-density polythene.
3. **Cross-linked Polymers:** They are composed of bifunctional and trifunctional monomers. They have a stronger covalent bond in comparison to other linear polymers. Bakelite and melamine are examples of cross-linked polymers.

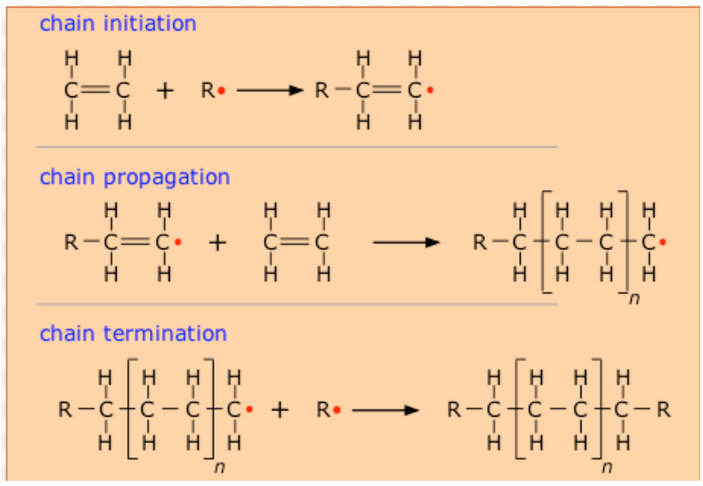
Classification Based on Molecular Forces

- **Elastomers:** These are rubber-like solids, and weak interaction forces are present in them. For example, rubber.
- **Fibres:** Strong, tough, high tensile strength and strong forces of interaction are present. For example, nylon -6, 6.
- **Thermoplastics:** These have intermediate forces of attraction. For example, polyvinyl chloride.
- **Thermosetting polymers:** These polymers greatly improve the material's mechanical properties. It provides enhanced chemical and heat resistance. For example, phenolics, epoxies and silicones.

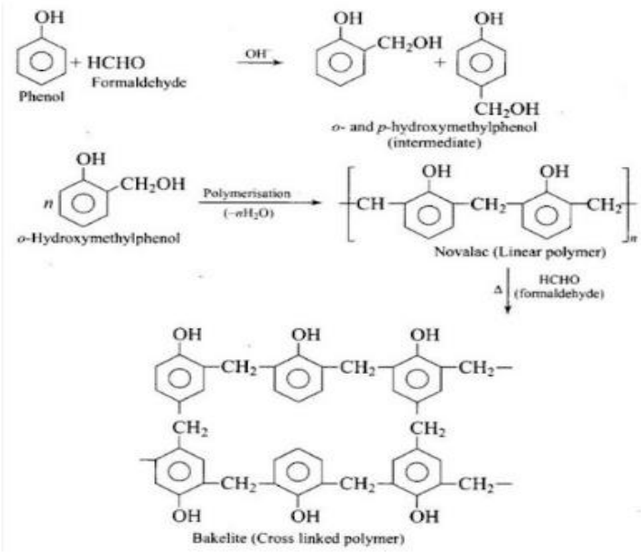
Thermosoftening plastics	**Thermosetting plastic**
Can be melted and remolded multiple times.	Cannot be remelted once cured; forms irreversible bonds.
Recyclable; can be reheated and reshaped.	Not recyclable; cannot be reshaped after curing.
Examples: Polyethylene (PE), Polypropylene (PP), Polyvinyl Chloride (PVC), Polystyrene (PS).	**Examples**: Epoxy resins, Phenolic resins, Polyurethane.
Used in plastic bags, bottles, toys, automotive parts.	Used in electronic housings, adhesives, coatings.
Less heat-resistant; can melt under high temperatures.	More heat-resistant; maintains shape under high temperatures.
Generally less durable and more flexible.	More durable and resistant to deformation.
Usually formed by addition polymerization	Usually formed by condensation polymerization
Contains long chain linear polymers, held together by weak Van der Waal forces	Contains 3-D network structure with strong covalent bonds
Soluble in organic solvents	Insoluble in organic solvents
Expensive	Cheap
Soft, weak, less brittle	Hard, strong, more brittle

Classification Based on Polymerization

1. **Addition Polymerization:** For example, polythene, Teflon, polyvinyl chloride (PVC), etc.



2. **Condensation Polymerization:** Examples include nylon -6, 6, perylene, polyesters, .



Addition Polymerization	**Condensation Polymerization**
Monomers must have double or triple bonds.	Monomers must be two similar or different functional groups.
Monomers add to each other without the loss of any small molecules.	Monomers join together with the elimination of small molecules like water, HCl, or ammonia.
No by-products are formed.	By-products such as water, HCl, or ammonia are formed.
Catalysts are often radical initiators or Lewis acids/bases.	Various catalysts can be used, depending on the specific reaction.
Examples: Polyethylene (PE), Polypropylene (PP), Polystyrene (PS), PVC.	Examples: Nylon, Polyester, Bakelite, Silicon.
The molecular weight of the polymer is a multiple of the monomer's molecular weight.	The molecular weight of the polymer is not a simple multiple of the monomer's molecular weight.

Applications of Polymers

Some important uses of polymers in our everyday life.

- Polypropene finds usage in a broad range of industries, such as textiles, packaging, stationery, plastics, aircraft, construction, rope, toys, etc.
- Polystyrene is one of the most common plastic actively used in the packaging industry. Bottles, toys, containers, trays, disposable glasses and plates, TV cabinets and lids are some of the daily-used products made up of polystyrene. It is also used as an insulator.
- The most important use of polyvinyl chloride is the manufacture of sewage pipes. It is also used as an insulator in electric cables.
- Polyvinyl chloride is used in clothing and furniture and has recently become popular for the construction of doors and windows as well. It is also used in vinyl flooring.
- Urea-formaldehyde resins are used for making adhesives, moulds, laminated sheets, unbreakable containers, etc.
- Glyptal is used for making paints, coatings and lacquers.

- Bakelite is used for making electrical switches, kitchen products, toys, jewellery, firearms, insulators, computer discs, etc.

Commercial Uses of Polymers

Polymer	Monomer	Uses of Polymer
Rubber	Isoprene (1, 2-methyl 1 – 1, 3-butadiene)	Making tyres, elastic materials
BUNA – S	(a) 1, 3-butadiene (b) Styrene	Synthetic rubber
BUNA – N	(a) 1, 3-butadiene (b) Vinyl Cyanide	Synthetic rubber
Teflon	Tetra Fluoro Ethane	Non-stick cookware – plastics
Terylene	(a) Ethylene glycol (b) Terephthalic acid	Fabric
Glyptal	(a) Ethylene glycol (b) Phthalic acid	Fabric
Bakelite	(a) Phenol (b) Formaldehyde	Plastic switches, Mugs, buckets
PVC	Vinyl Cyanide	Tubes, Pipes
Melamine Formaldehyde Resin	(a) Melamine (b) Formaldehyde	Ceramic, plastic material
Nylon-6	Caprolactam	Fabric

Disposal of Polymers

Plastics are **synthetic** polymers, which are widely used but they are difficult to dispose: They either are non-biodegradable or if they are, they take very long to break down.

There are different ways of disposal of polymers:

1. **Landfills –** Landfilling is a common method for disposing of polymer waste. It involves burying the waste in designated landfill sites. Plastics which are difficult to sort or recycle or that aren't plentiful enough to separate, are usually buried in a landfill.

   Advantages
   - **Cost-Effective**: Landfilling is generally less expensive compared to other disposal methods like incineration or recycling
   - **Simplicity**: The process is straightforward and does not require complex technology or infrastructure
   - **Capacity**: Landfills can handle large volumes of waste, making them suitable for disposing of significant amounts of polymer waste
   - **Immediate Solution**: Provides an immediate solution for waste management, especially in areas lacking recycling facilities

   Disadvantages
   - **Environmental Impact**: Polymers in landfills can take hundreds of years to decompose, leading to long-term environmental pollution
   - **Space Consumption**: Landfills require large areas of land, which can lead to land use issues and habitat destruction
   - **Leachate Production**: Decomposing waste can produce leachate, a liquid that can contaminate soil and groundwater if not properly managed
   - **Greenhouse Gas Emissions**: Landfills can emit methane, a potent greenhouse gas, contributing to climate change
   - **Non-Renewable**: Landfilling does not recover any material value from the polymers, unlike recycling

2. **Incineration –** Incineration involves burning polymer waste at high temperatures, significantly reducing its volume and potentially generating energy in the form of heat or electricity. This method is effective for managing large amounts of waste and can destroy hazardous substances, reducing environmental risks.

   Advantages
   - **Volume Reduction**: Incineration significantly reduces the volume of waste, which helps in managing landfill space more efficiently
   - **Energy Generation**: The process can generate energy in the form of heat or electricity, which can be harnessed for various uses

- **Hazardous Waste Destruction**: Incineration can effectively destroy hazardous substances in polymers, reducing the risk of environmental contamination
- **Immediate Disposal**: Provides a quick solution for disposing of large amounts of polymer waste

Disadvantages

- **Air Pollution**: Incineration can release harmful pollutants, including dioxins and furans, into the atmosphere, contributing to air quality issues
- **High Costs**: The setup and operation of incineration plants can be expensive, requiring significant investment in technology and infrastructure
- **Energy Inefficiency**: While energy is generated, the overall efficiency can be lower compared to other energy recovery methods
- **Loss of Material**: Incineration does not allow for the recovery of materials, unlike recycling, which means the material value of polymers is lost

3. **Recycling –** Recycling is a sustainable method for managing polymer waste by converting it back into raw materials for new products. There are several key recycling methods:

   - **Mechanical Recycling**: This involves physically processing polymers through steps like grinding, washing, and re-melting. It's commonly used for plastics like PET and HDPE, but the quality of the recycled material can degrade over time
   - **Chemical Recycling**: This method breaks down polymers into their monomers or other basic chemicals using processes like pyrolysis, gasification, or depolymerization. It can handle a wider variety of plastics and produce high-quality raw materials
   - **Energy Recovery**: Also known as quaternary recycling, this process involves incinerating polymer waste to generate energy. While it reduces waste volume and produces energy, it can also release pollutants if not properly managed

   Recycling helps reduce the environmental impact of polymer waste by conserving resources, reducing landfill use, and lowering greenhouse gas emissions. However, it requires proper sorting and processing to be effective.

   Advantages

   - **Resource Conservation**: Recycling polymers conserves natural resources by reducing the need for raw materials
   - **Energy Savings**: It generally requires less energy to recycle polymers than to produce new ones from raw materials
   - **Waste Reduction**: Recycling reduces the amount of waste sent to landfills and incinerators, helping to manage waste more effectively
   - **Environmental Protection**: By reducing the need for new raw materials and minimizing waste, recycling helps lower greenhouse gas emissions and pollution
   - **Economic Benefits**: Recycling can create jobs and stimulate economic growth through the recycling industry

   Disadvantages

   - **Quality Degradation**: Recycled polymers can degrade in quality over time, limiting their use in certain applications
   - **Contamination Issues**: Effective recycling requires proper sorting and cleaning of materials, which can be challenging and costly
   - **Limited Recycling Infrastructure**: Not all regions have the necessary infrastructure to support efficient recycling programs
   - **Economic Viability**: The process can be economically unviable if the cost of recycling exceeds the value of the recycled materials
   - **Energy and Resource Use**: While recycling saves energy compared to producing new materials, it still requires energy and resources for collection, transportation, and processing

4. **Biodegradation**: breaking down the polymer into natural substances through the action of microorganisms.

Advantages

- **Environmental Benefits:** Biodegradation breaks down polymers into natural substances, reducing long-term environmental pollution and minimizing the impact on ecosystems
- **Reduction of Landfill Use**: Biodegradable polymers decompose naturally, reducing the volume of waste that ends up in landfills
- **Lower Greenhouse Gas Emissions:** Properly managed biodegradation can result in lower greenhouse gas emissions compared to incineration or landfilling
- **Sustainable Cycle:** Biodegradable polymers can be part of a sustainable lifecycle, returning to the environment without leaving harmful residues

Disadvantages

- **Slow Decomposition:** The rate of biodegradation can be slow and depends on environmental conditions such as temperature, moisture, and microbial activity
- **Limited Applications**: Not all polymers are biodegradable, and those that are may not be suitable for all applications due to their physical properties
- **Contamination Risk:** Biodegradable polymers can still cause environmental harm if they contain additives or contaminants that do not break down
- **Higher Costs:** The production and processing of biodegradable polymers can be more expensive compared to conventional plastics

The efficiency of each method of polymer disposal varies and depends on several factors such as the type of polymer, the disposal method used, and the conditions under which the disposal takes place. For example, recycling is the most sustainable method, but it may not always be possible for certain types of polymers. Incineration is efficient in terms of volume reduction, but it can release harmful gases into the atmosphere.

Environmental impacts of Polymer Disposal

Incorrect or inadequate polymer disposal can have significant environmental impacts, such as:

- Release of harmful chemicals into the air, water, and soil.
- Contamination of soil and water resources.
- Accumulation of plastic waste in landfills and the natural environment.
- Negative impacts on wildlife and ecosystems.

How to reduce the environmental impact of Polymer Disposal

We can reduce the environmental impact of polymer disposal by:

- Properly sorting and recycling polymers.
- Minimizing the use of single-use plastic items.
- Properly disposing of polymers in designated landfill sites.
- Promoting the use of biodegradable polymers and alternative materials.

Properties of polymers

Solubility: Solubility of polymers, that is, ability to dissolve spontaneously in low-molecular-weight solvent and to form thermodynamically stable molecular solutions.

Solubility depends on the nature of the solvent, the temperature of the solution, the topology of the polymer, and the crystallinity. Cross-linked polymers do not dissolve, but only swell with the solvent when an interaction between the polymer and the solvent occurs.

Polymer solubility is an important consideration in the coatings and adhesives industries. Most methods of polymer analysis are based on measurements of their solution, such as light scattering, osmometry, gel-permeation chromatography, and viscometry.

Factors affecting solubility of polymer:

- Polarity of the solvent: interaction between polymer chain and solvent molecules.
- Intermolecular forces: strong dipole-dipole or hydrogen bonding interactions hinders the penetration of solvent molecules between polymeric chains.
- Molecular weight: higher the MW-difficult to solvate.
- Structure: Long chains can be densely entangled. This makes it difficult for solvent molecules to penetrate. So longer chains dissolve more slowly than short ones. As crosslinking increases, solubility decreases.

Molecular Weight: A polymer's molecular weight is the sum of the atomic weights of individual atoms that comprise a molecule. It indicates the average length of the bulk resin's polymer chains. Not all polymer molecules of a particular grade have the exact same molecular weight. There is a range or distribution of molecular weights.

Low Molecular Weight polymers have lower properties due to lack of chain entanglement and higher molecular mobility. Shorter chains have more end groups which increase free volume and molecular mobility which makes it easier for the polymer chains to respond to stresses.

MW of plastics influences their properties. With increasing MW, polymer properties increase for abrasion resistance, brittleness, chemical resistance, elongation, hardness, melt viscosity, tensile strength, modulus, toughness, and yield strength.

There are two types of average molecular masses of polymers.

- **Number Average Molecular Masses**
- **Weight Average Molecular Mass**

Number Average Molecular Masses

If N_1, N_2, N_3.... are the number of macromolecular with molecular masses. M_1, M_2, M_3....., respectively, then the number average molecular masses of the polymer is given by

$$\bar{M}_n = \frac{N_1M_1+N_2M_2+N_3M_3+\ldots..\sum N_iM_i}{N_1+N_2+N_3+\ldots..\sum N_i}$$

The number average molecular mass

$\bar{M}_n$

is determined by colligative properties such as osmotic pressure.

Weight Average Molecular Mass

If m_1, m_2, m_3.... are the masses of a macromolecule with molecular masses M_1, M_2, M_3..., respectively, then the weight average molecular mass of the polymer is given by

$$\overline{M}_\omega = \frac{m_1M_1+m_2M_2+m_3M_3+\ldots\ldots}{m_1+m_2+m_3}$$
$$= \frac{\sum miMi}{\sum mi}$$
$$\Rightarrow \overline{M}_\omega = \frac{\sum NiMi \times Mi}{\sum NiMi}$$
$$\Rightarrow \overline{M}_\omega = \frac{\sum NiMi^2}{\sum NiMi}$$

Polydispersive index: It is the ratio of the weight average molecular mass and the number average molecular mass of polymers.

$$PDI = \frac{\bar{M}w}{\bar{M}n}$$

. For natural polymers, PDI = 1.

Crystallinity: A polymer is said to be crystalline if all the molecules are arranged in an orderly manner with symmetrical orientation with higher force of attraction between two chains.

Any polymer will contain definite percentage of crystalline part and amorphous part. Crystalline parts are formed when individual chains are linear, contain no bulky substituents and are closely arranged parallel to each other. The chains are bonded together by van der waal's forces or hydrogen bonding or polar interaction.

Eg. HDPE, PVC are highly crystalline whereas polypropylene, polystyrene are less crystalline.

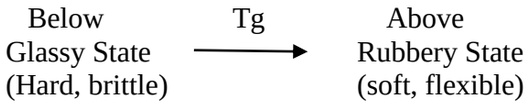
Properties of Crystalline polymers:

- High tensile strength
- High density.
- High impact and wear resistance
- High Tg
- Hard but brittle and strong
- Solubility and permeability decreases with increase in crystallinity

Glass transition temperature. The glass transition temperature (Tg) is defined as the temperature below which an amorphous (having no definite shape, form or structure.) polymer is brittle and glassy and above that temperature, it becomes flexible, soft and rubbery.

At this temperature, polymers undergo a transition from glassy to rubbery state. Tg is an important feature of polymer behaviour. It marks a region of dramatic changes in the physical and mechanical properties.

- **Below Tg**: Due to lack of mobility, the polymers are hard and brittle like glass.
- **Above Tg**: Due to some mobility, the polymers are soft and flexible like rubber.



The units of glass transition temperature are:

- Degree Celsius (°C)
- Degree Fahrenheit (°F)
- Kelvin (K)

The value depends on the mobility of the polymer chain. The Tg for most synthetic polymers lies between 170K to 500K.

Polymers with high Tg: Polystyrene, Poly(methyl methacrylate)
These polymers are hard and brittle. Their Tg's are higher than room temperature.

Polymers with low Tg: Rubber elastomers like Polyisoprene, Polyisobutylene
They are soft and flexible in nature. Their Tg's are less than room temperature.

Factors affecting value of Tg

1. Crystallinity: Crystalline polymers have higher Tg than amorphous polymers because in crystalline polymers, polymer chains are arranged in regular fashion and each chain is bound to each other by strong bonds like H-bonding.
2. Molecular Weight: Tg of polymer increases with molecular weight up to a particular value. Beyond that value, there is no change.
3. Tg increases with presence of side chains as it hinders free rotation.
4. Presence of plasticizers reduces Tg of polymer.